

Performing a Non-credential Scan Using Nessus

Objective: To perform a non-credential scan using Nessus to identify potential vulnerabilities without requiring login credentials

Tools required: Windows Server 2022 and Metasploitable

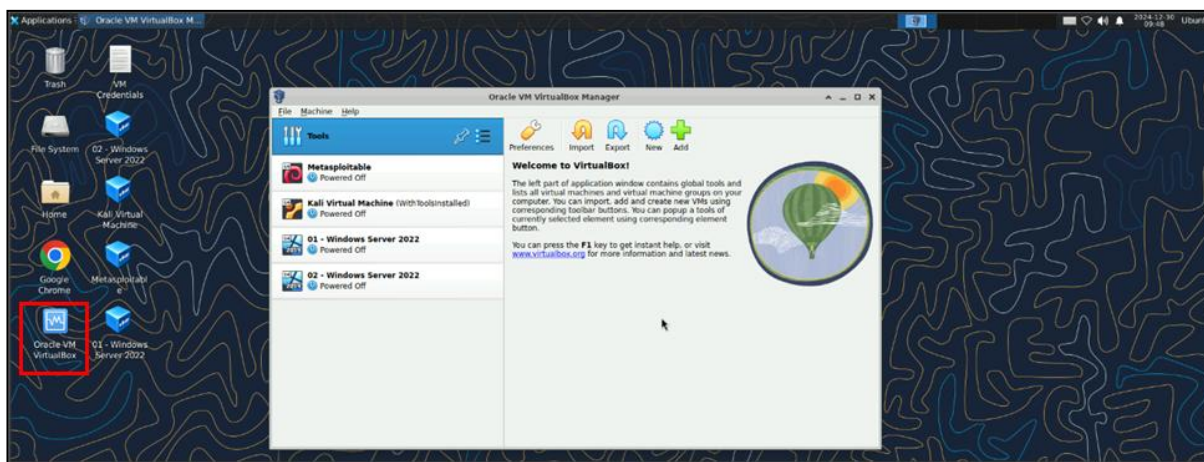
Prerequisites: Nessus must be installed and configured.

Steps to be followed:

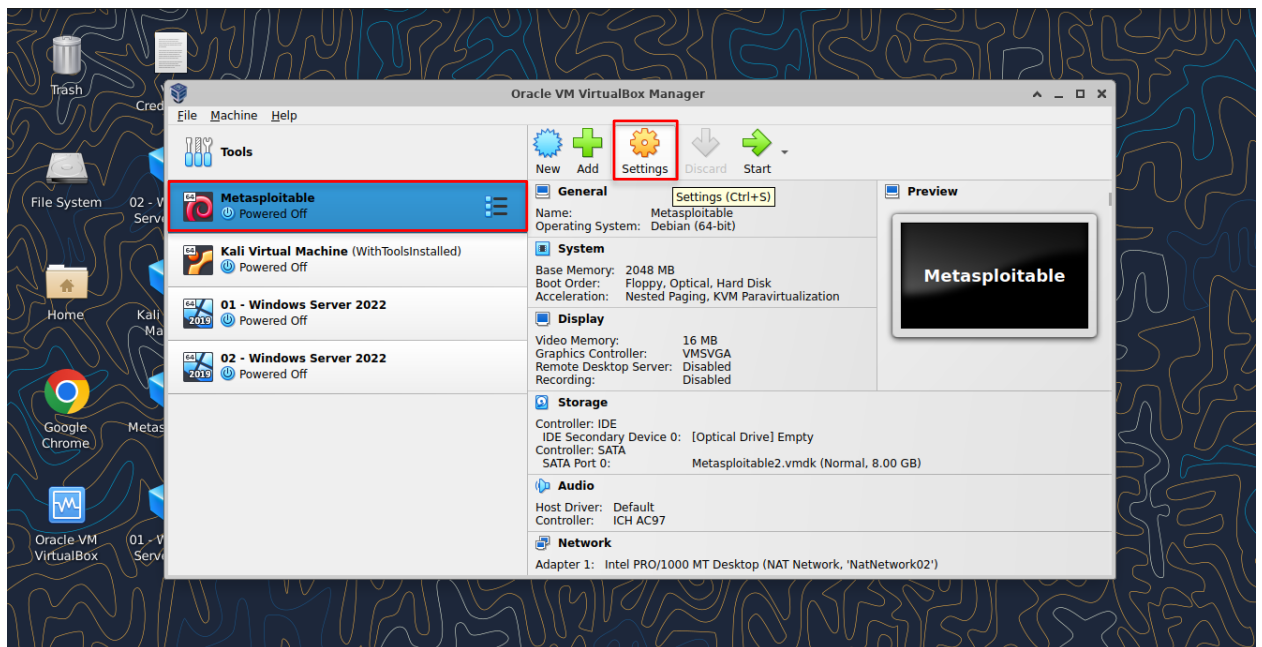
1. Set up the network
2. Perform a non-credential scan

Step 1: Set up the network

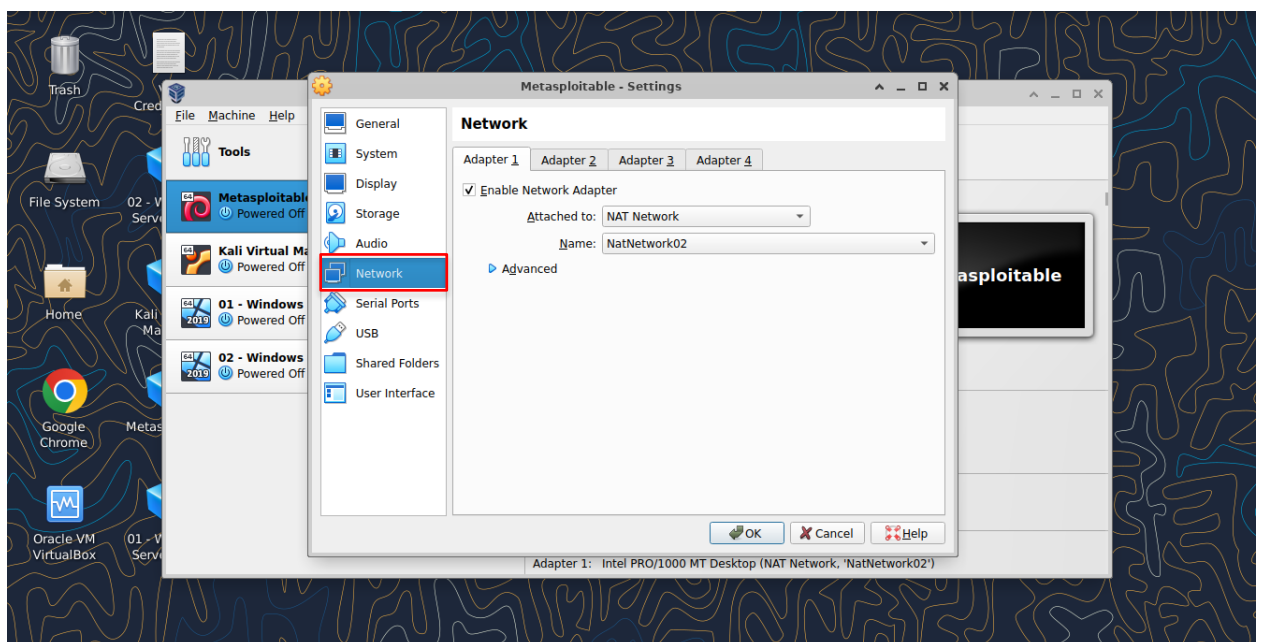
1.1 Open **Oracle VM VirtualBox Manager** by double-clicking on it



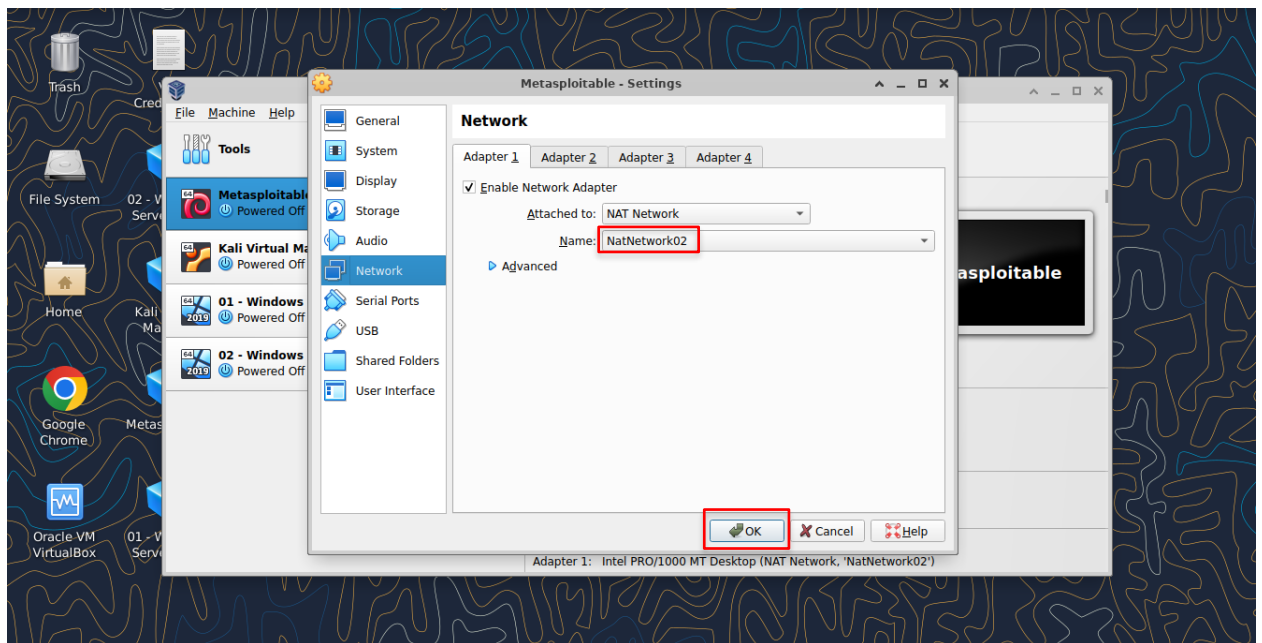
1.2 Select **Metasploitable**, then click **Settings**



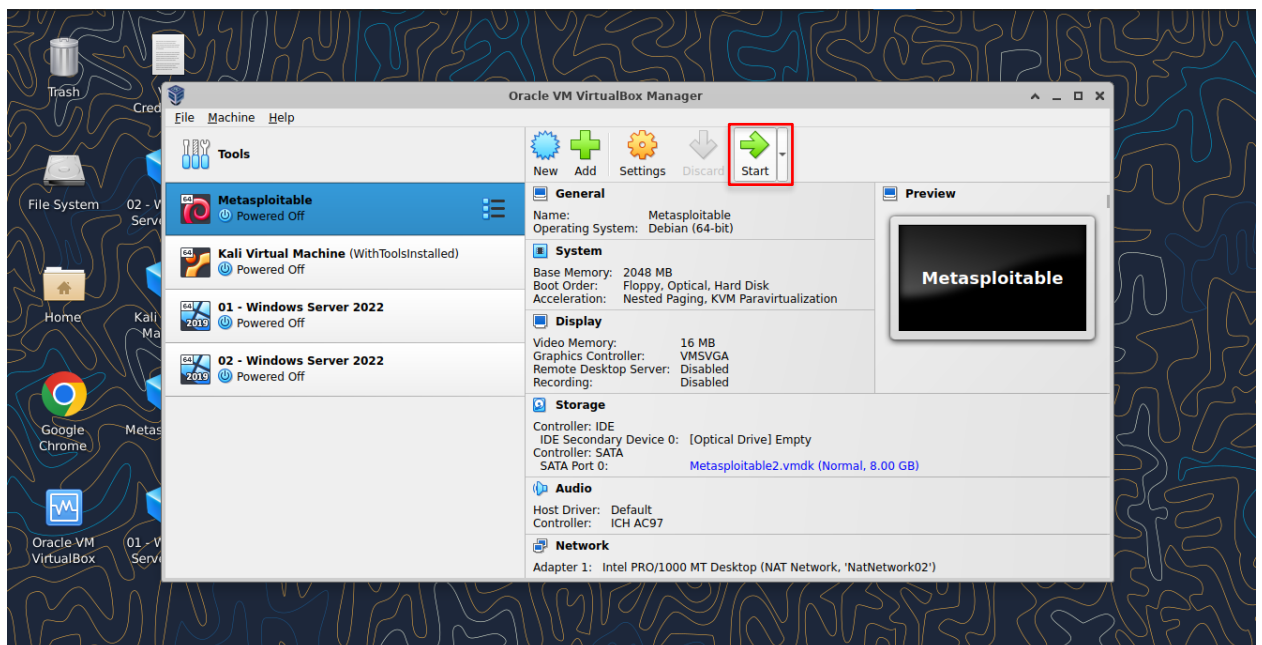
1.3 Click the **Network** tab



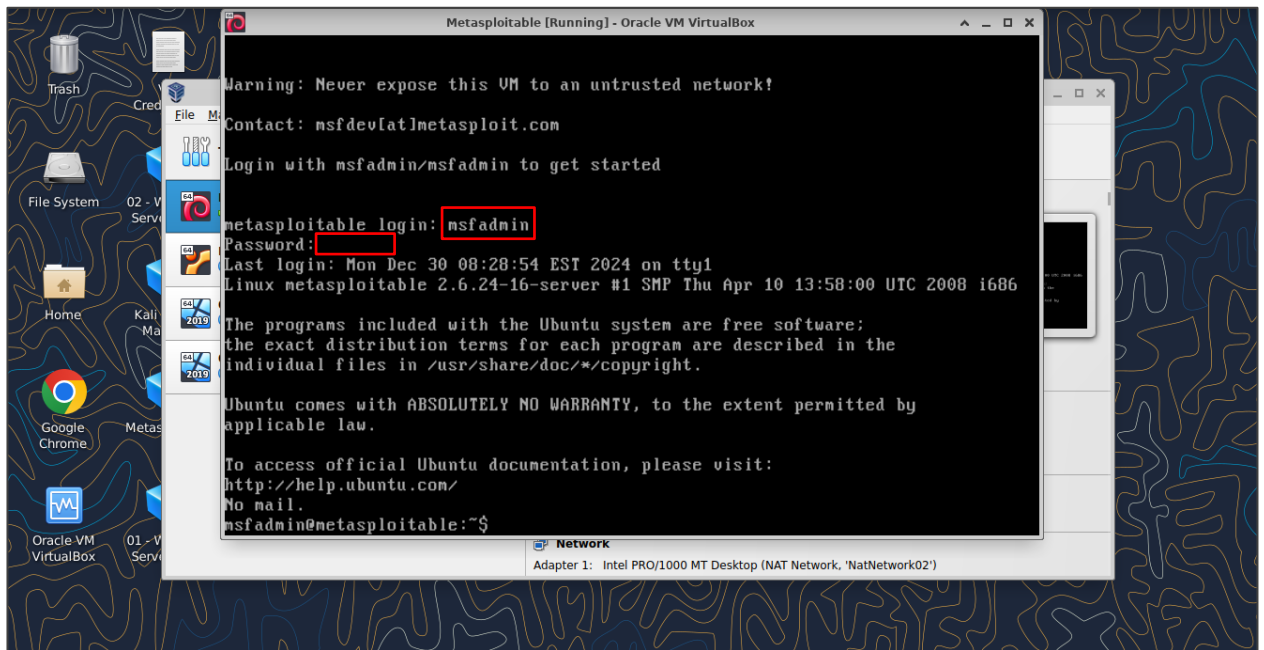
1.4 Select **NatNetwork02**, then click **OK**



1.5 Click **Start** to power on the Metasploitable machine

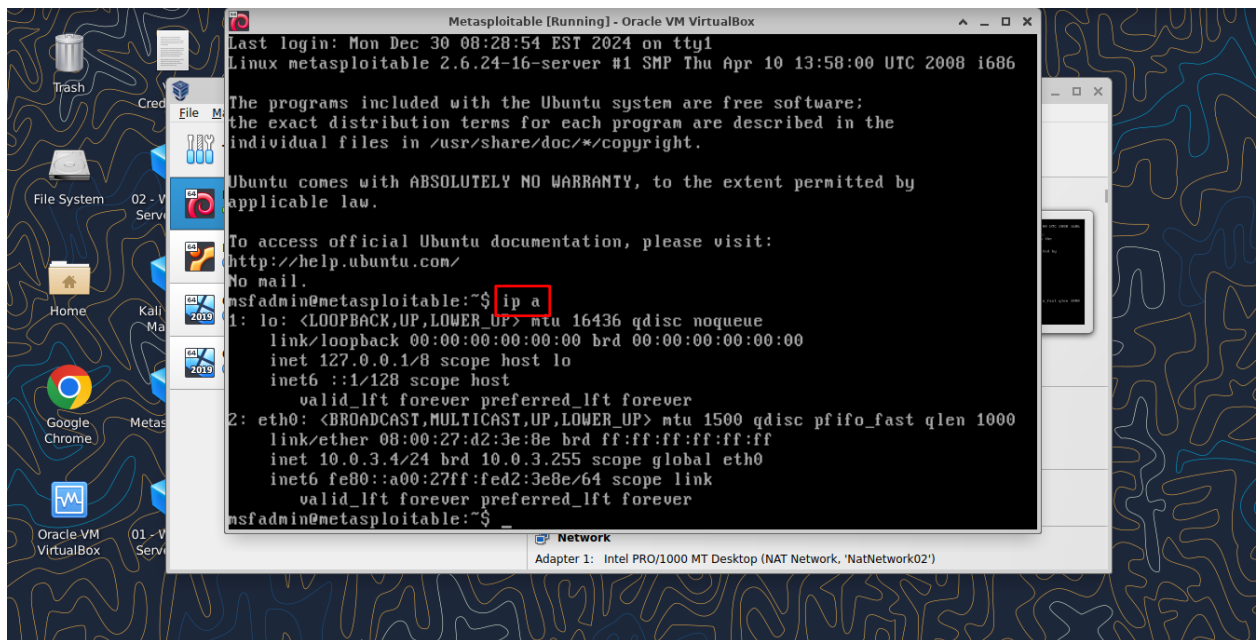


1.6 Enter **msfadmin** as both username and password for metasploitable login

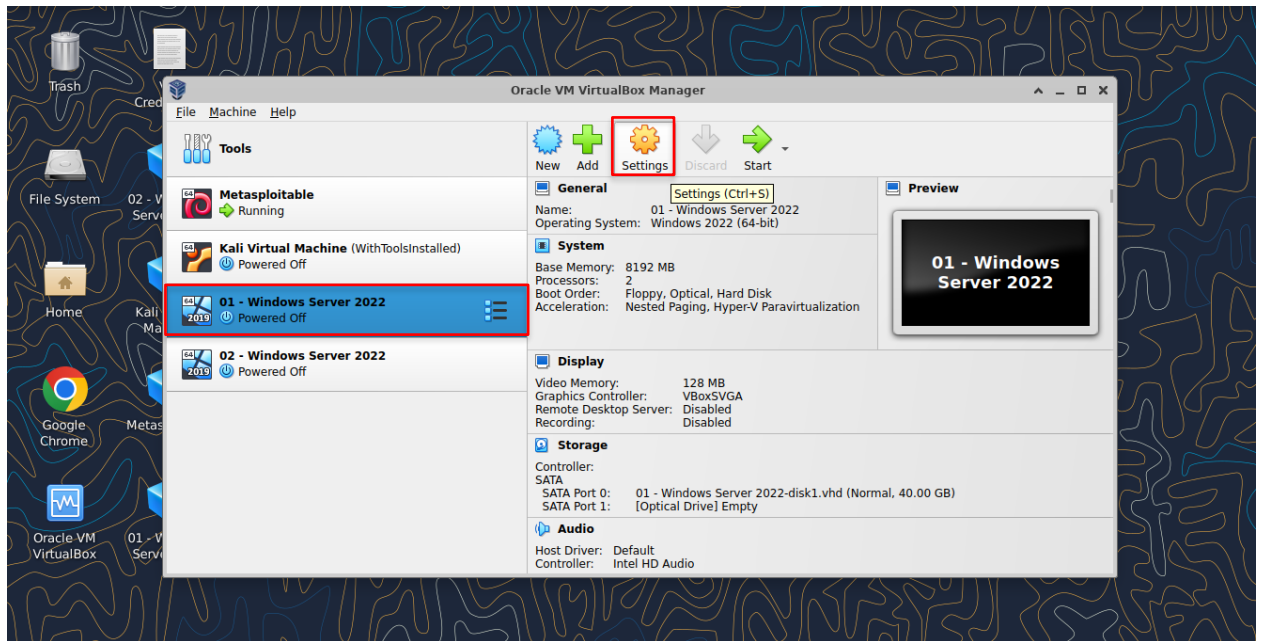


1.7 Run the following command to check the IP address:

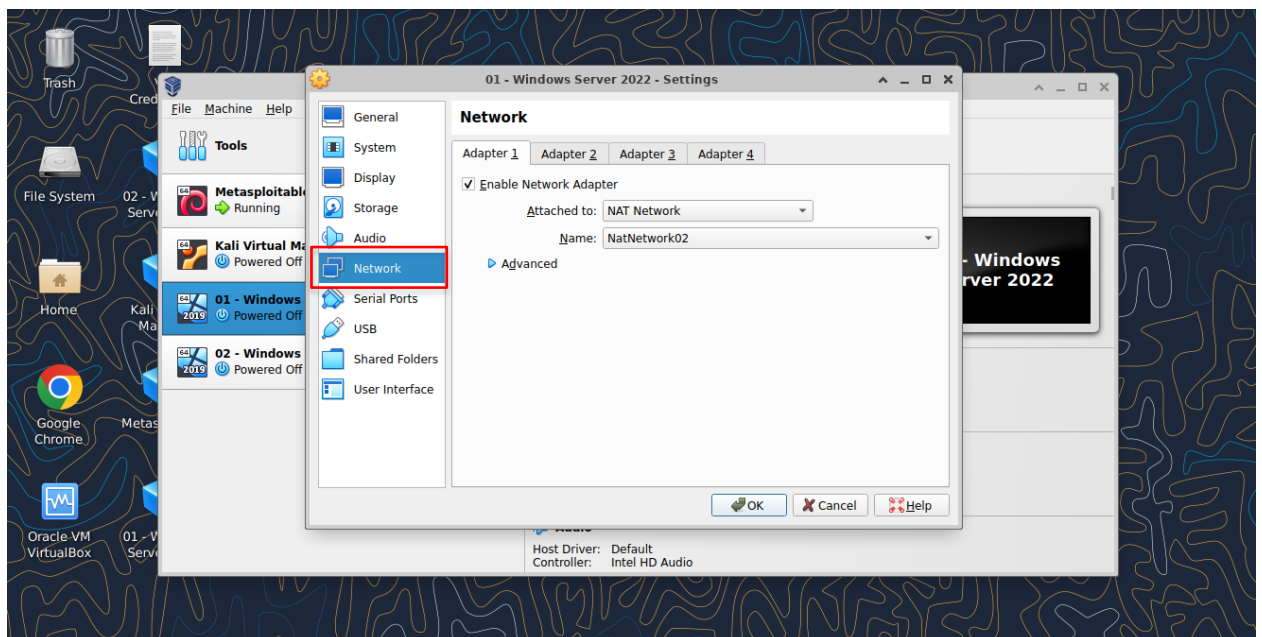
ip a



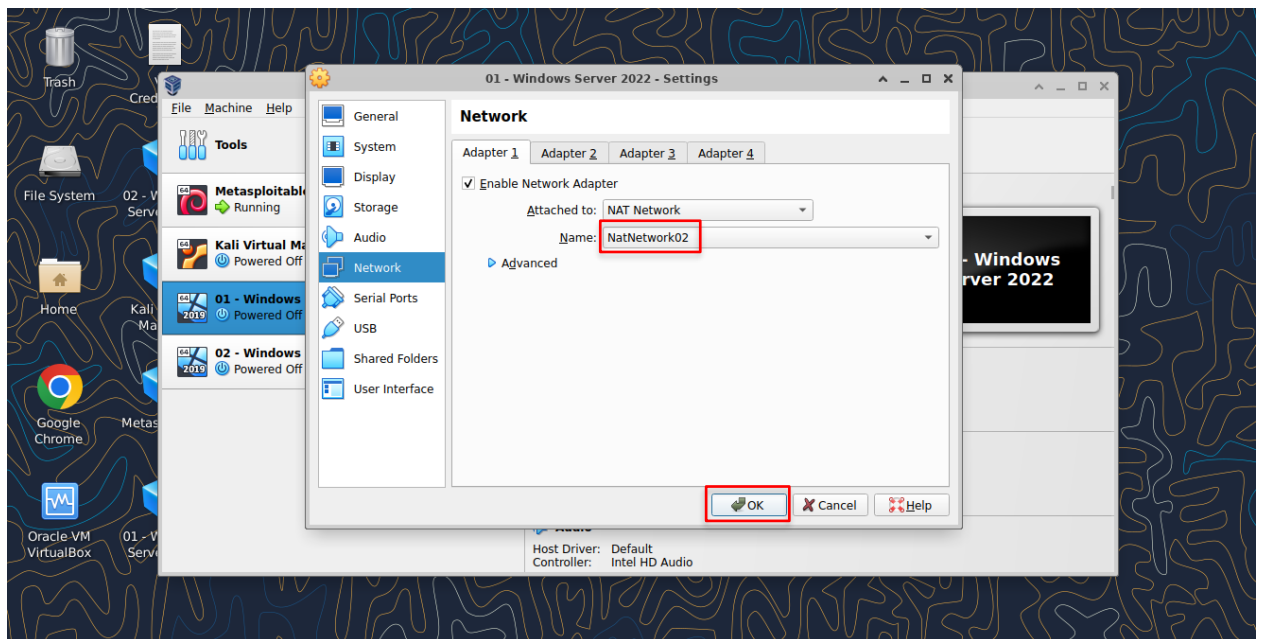
1.8 Select **01 - Windows Server 2022** in Oracle VirtualBox and click **Settings**



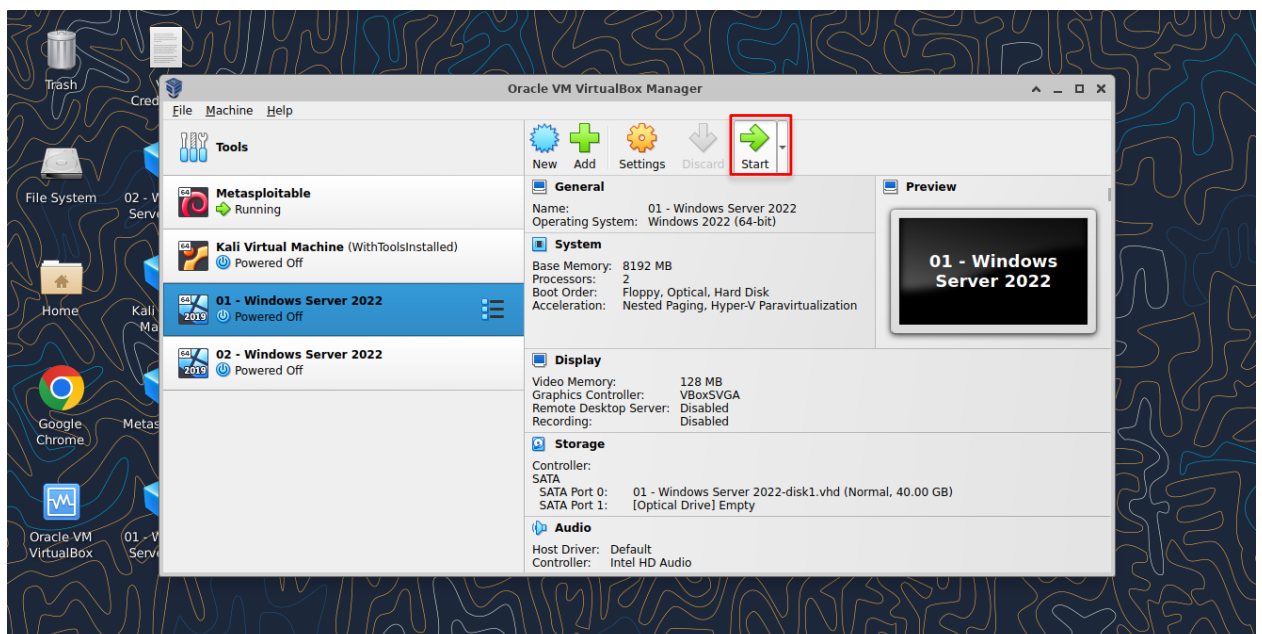
1.9 Click the **Network** tab



1.10 Select **NatNetwork02**, then click **OK**

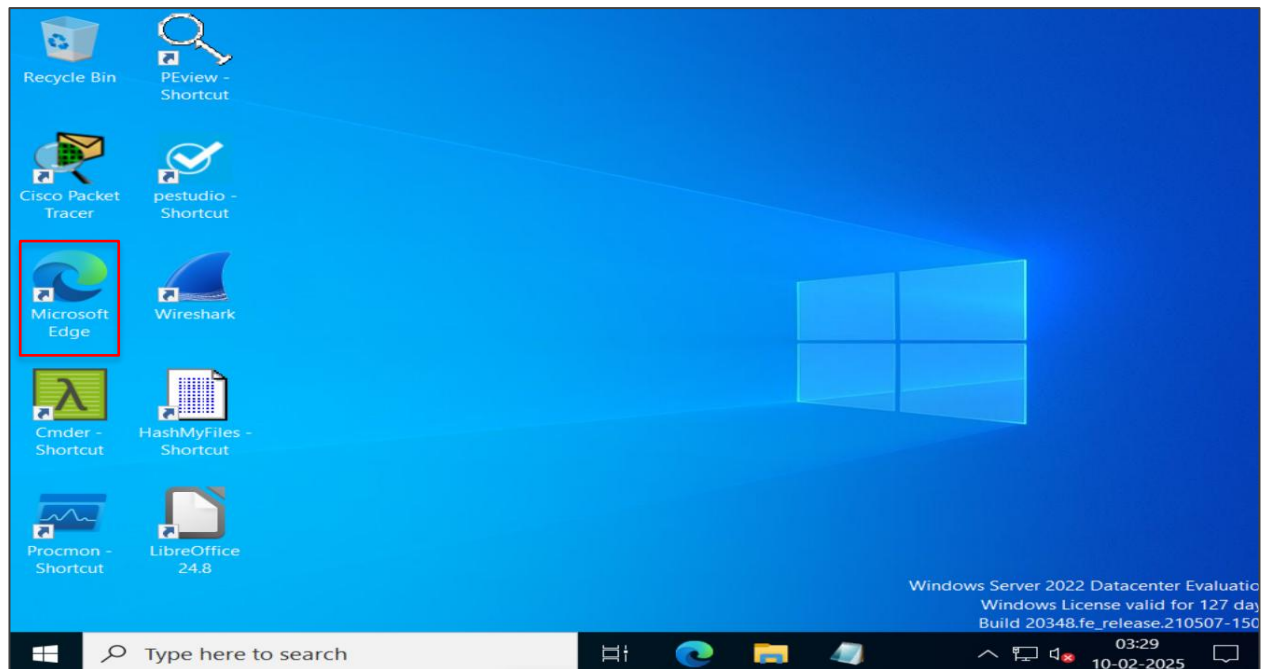


1.11 Click **Start** to power on Windows Server 2022



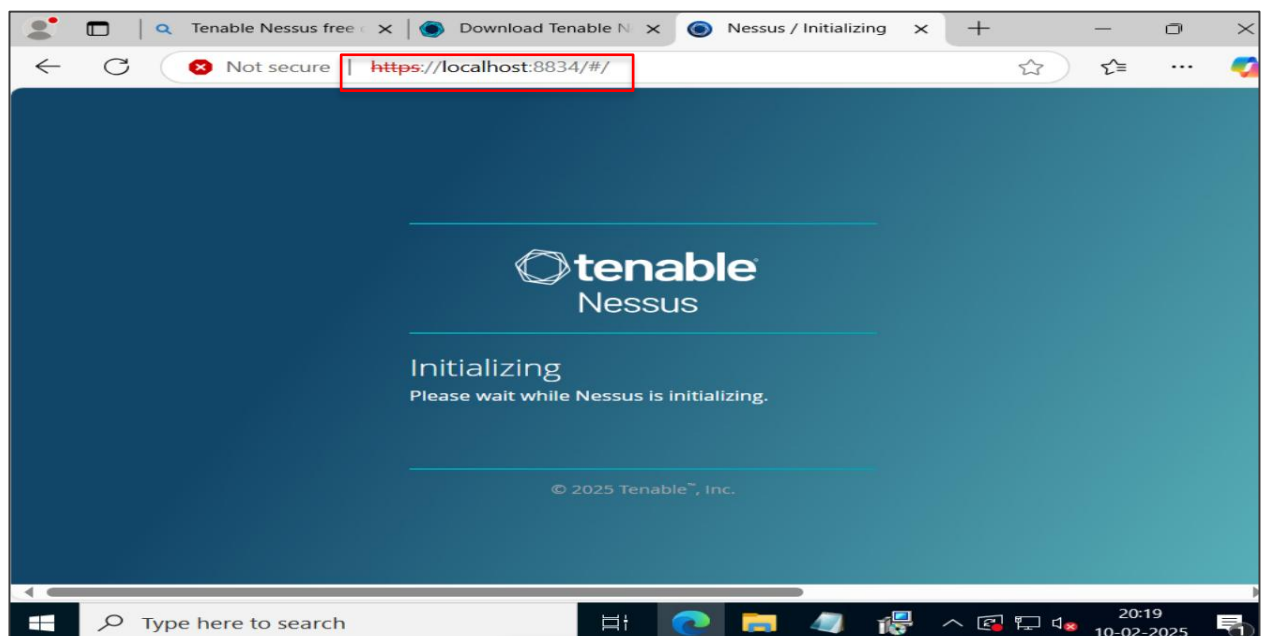
Step 2: Perform a non-credential scan

2.1 Open Microsoft Edge by clicking on it

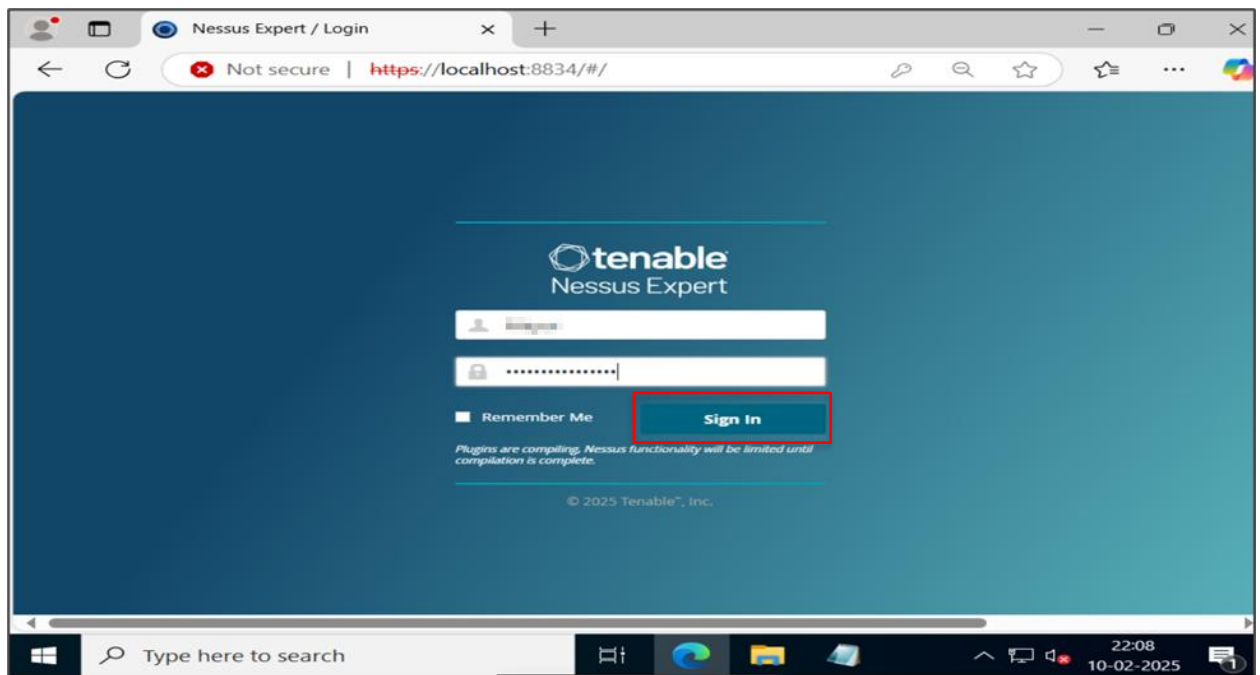


2.2 Navigate to the following URL:

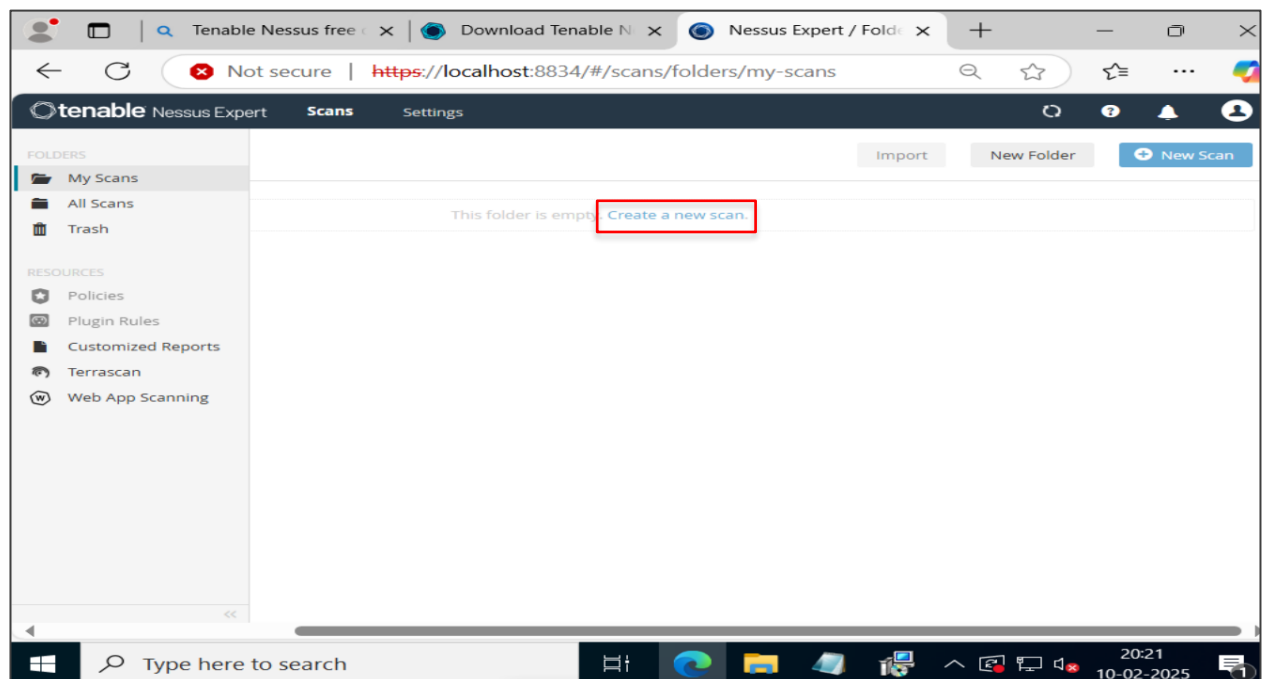
<https://localhost:8834/#/>



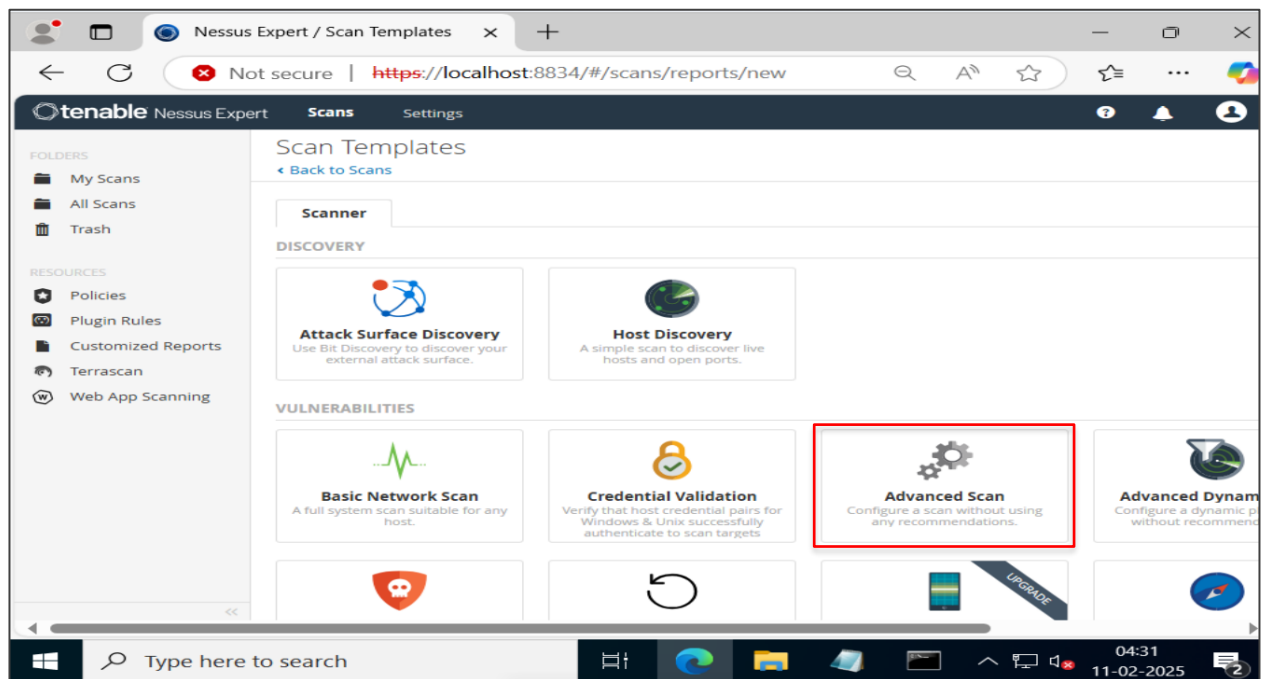
2.3 Enter your Nessus credentials and click **Sign In**



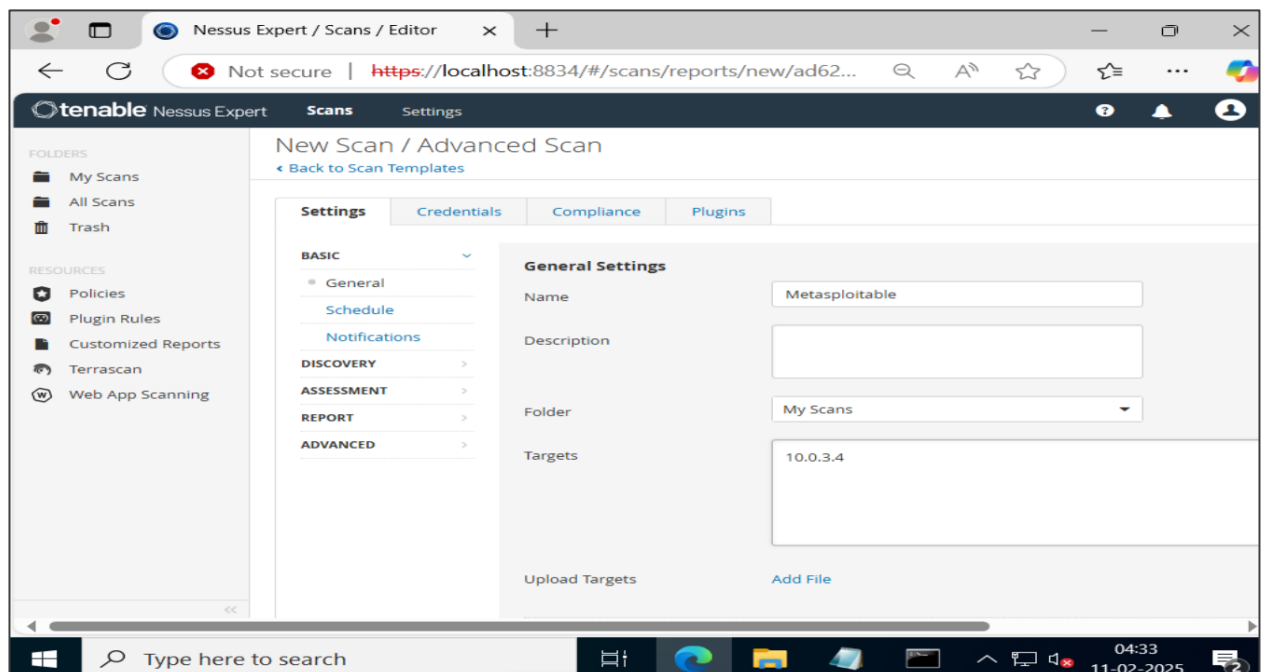
2.4 Click **Create a new scan**



2.5 Select Advanced Scan

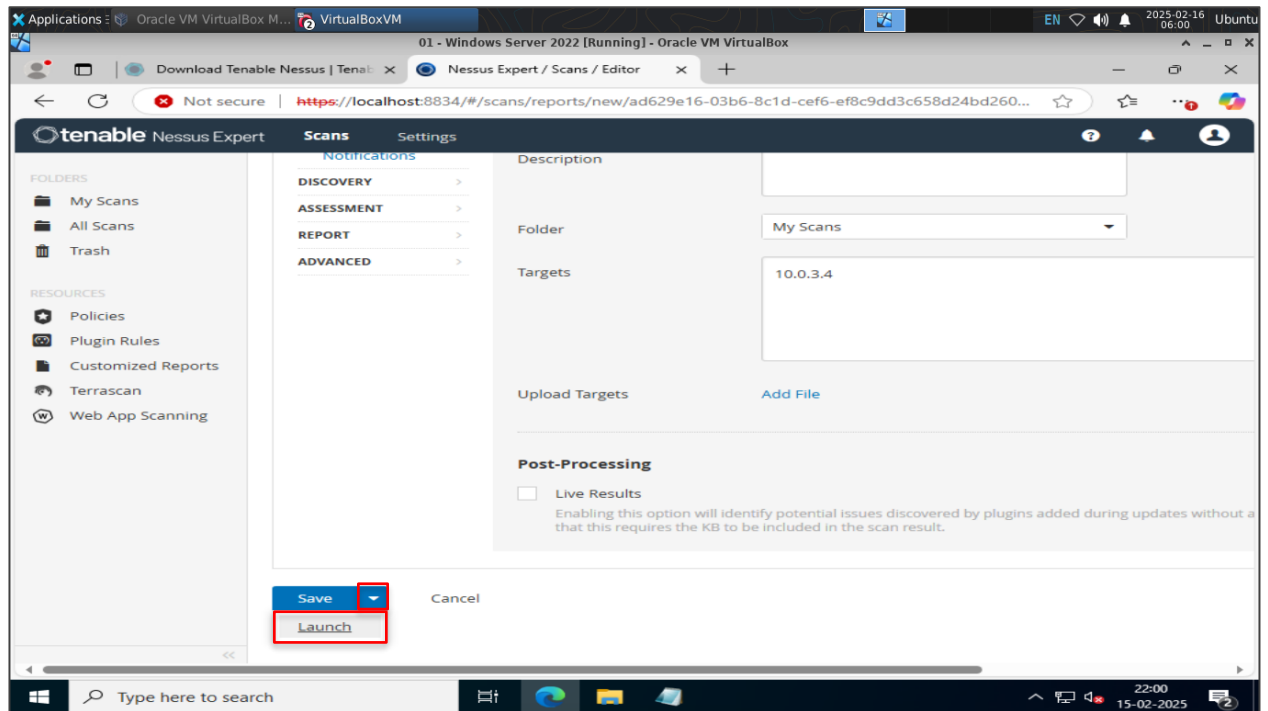


2.6 Enter Metasploitable as Name and 10.0.3.4 as Targets

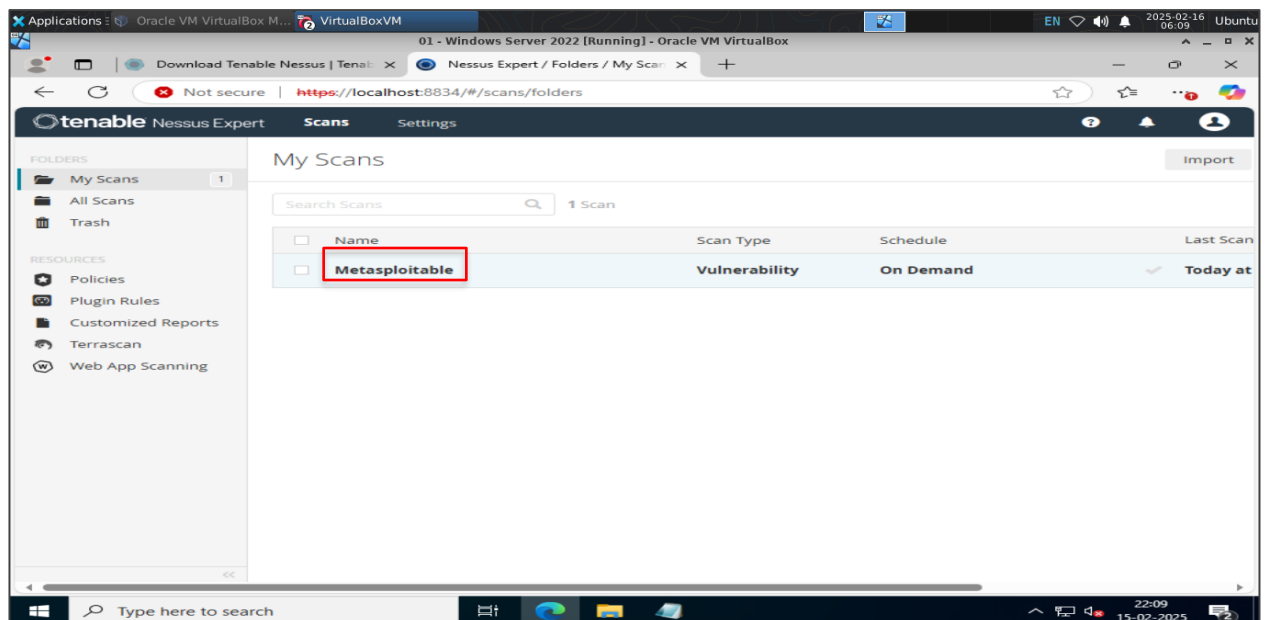


Note: Replace **10.0.3.4** with the actual IP address of your Metasploitable machine

2.7 Click the drop-down menu and select **Launch**



2.8 Click **Metasploitable** to open the scan results



You can now view the complete scan summary.

The screenshot shows the Nessus Expert interface for a scan named 'Metasploitable'. The interface includes a sidebar with navigation options like 'My Scans', 'All Scans', and 'Trash'. The main content area displays the 'Scan Summary' with various metrics and details.

Scan Details:

- Scan Name: Metasploitable
- Plugin Set: 202502160144
- CVSS_Score: CVSS_V3
- Scan Template: Advanced Scan
- Scan Start: Today at 10:00 PM
- Scan End: Today at 10:06 PM

Authentication / Credential Info (Hosts):

Hosts	Authentication / Credential Info (Hosts)
0	SUCCEEDED
1	FAILED

Top 5 Operating Systems Detected During Scan:

- AIX 5.3

Scan Durations:

Scan Duration	Median Scan Time Per Host	Max Scan Time
00:05:54	00:05:54	00:05:54

Plugin Families Enabled/Disabled:

Search Plugin Families: 61 Plugin Families

Status: disabled Family Name: Mobile Devices

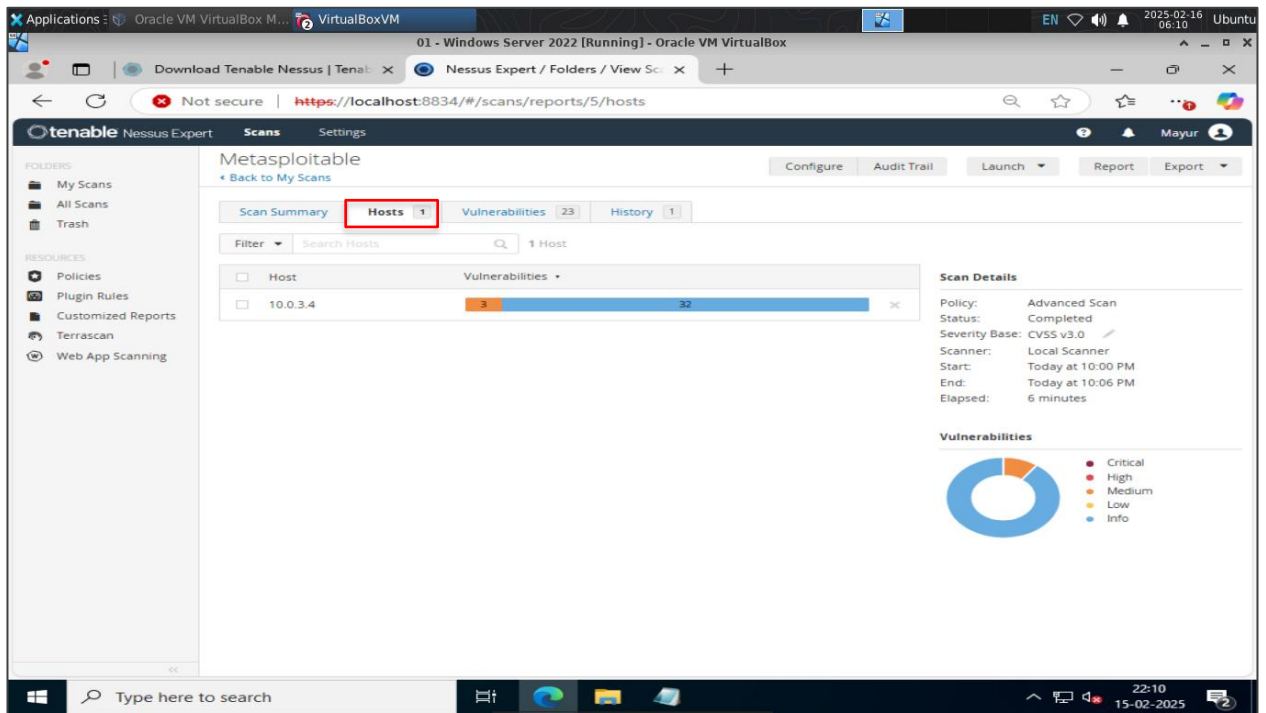
2.9 Scroll down to view the scan status and vulnerability categories

The screenshot shows the Nessus Expert interface with the 'Scan Summary' page. The 'Plugin Families Enabled/Disabled' section is highlighted with a red box, showing a list of plugin families and their status.

Status	Family Name
disabled	Mobile Devices
enabled	SMTP problems
enabled	Rocky Linux Local Security Checks
enabled	Backdoors
enabled	Ubuntu Local Security Checks
enabled	RPC
enabled	Gentoo Local Security Checks
enabled	Oracle Linux Local Security Checks
enabled	Artificial Intelligence
enabled	Brute force attacks

Results per page: 10

2.10 Click **Hosts** to view host details



The screenshot shows the Tenable Nessus Expert web interface. The browser address bar indicates the URL <https://localhost:8834/#/scans/reports/5/hosts>. The interface displays a scan named 'Metasploitable'. The 'Hosts' tab is selected and highlighted with a red box. The 'Hosts' tab shows a table with one host, '10.0.3.4', which has 32 vulnerabilities. The 'Vulnerabilities' tab shows 23 vulnerabilities. The 'Scan Details' section on the right provides information about the scan, including the policy, status, severity base, scanner, start and end times, and elapsed time. A donut chart shows the distribution of vulnerability severity levels: Critical (0), High (0), Medium (0), Low (0), and Info (32).

Host	Vulnerabilities
10.0.3.4	32

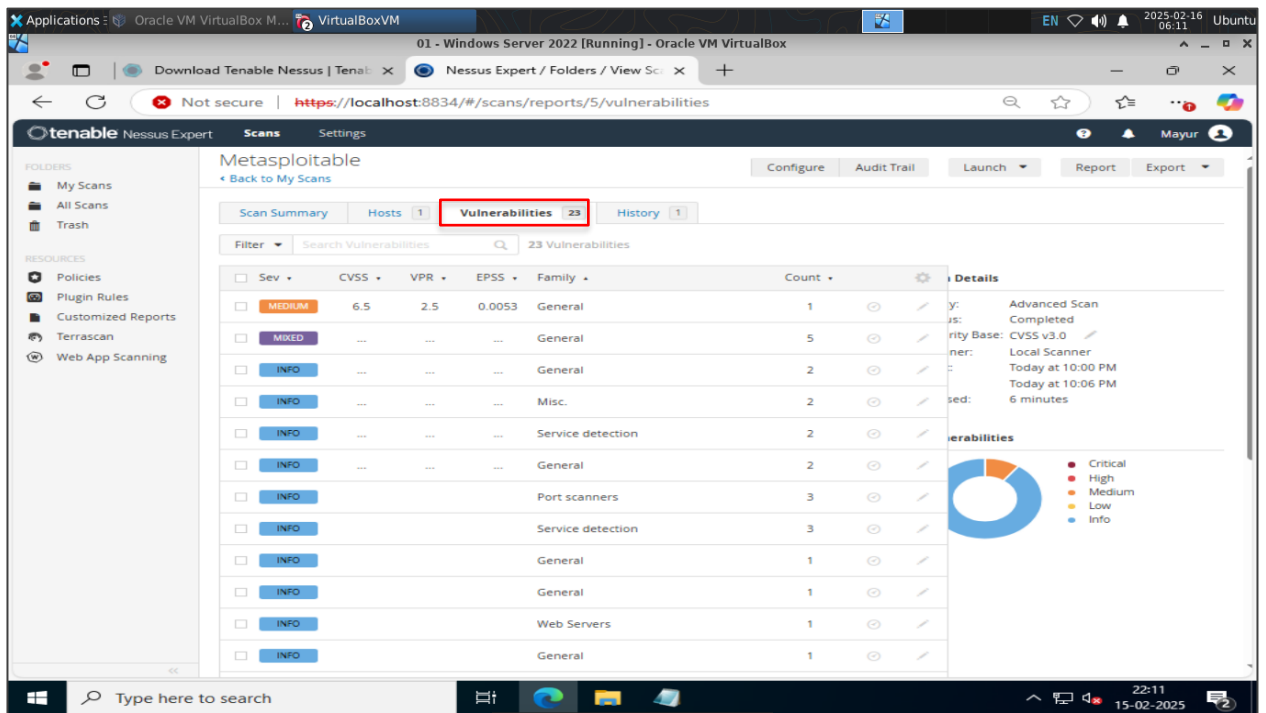
Scan Details

- Policy: Advanced Scan
- Status: Completed
- Severity Base: CVSS v3.0
- Scanner: Local Scanner
- Start: Today at 10:00 PM
- End: Today at 10:06 PM
- Elapsed: 6 minutes

Vulnerabilities

- Critical
- High
- Medium
- Low
- Info

2.11 Click **Vulnerabilities** to see information about detected security issues



The screenshot shows the Tenable Nessus Expert web interface. The browser address bar indicates the URL <https://localhost:8834/#/scans/reports/5/vulnerabilities>. The interface displays a scan named 'Metasploitable'. The 'Vulnerabilities' tab is selected and highlighted with a red box. The 'Vulnerabilities' tab shows a table with 23 vulnerabilities. The 'Scan Details' section on the right provides information about the scan, including the policy, status, severity base, scanner, start and end times, and elapsed time. A donut chart shows the distribution of vulnerability severity levels: Critical (0), High (0), Medium (0), Low (0), and Info (32).

Sev	CVSS	VPR	EPSS	Family	Count
MEDIUM	6.5	2.5	0.0053	General	1
MIXED	...	...	...	General	5
INFO	...	...	...	General	2
INFO	...	...	...	Misc.	2
INFO	...	...	...	Service detection	2
INFO	...	...	...	General	2
INFO	...	...	...	Port scanners	3
INFO	...	...	...	Service detection	3
INFO	...	...	...	General	1
INFO	...	...	...	General	1
INFO	...	...	...	Web Servers	1
INFO	...	...	...	General	1

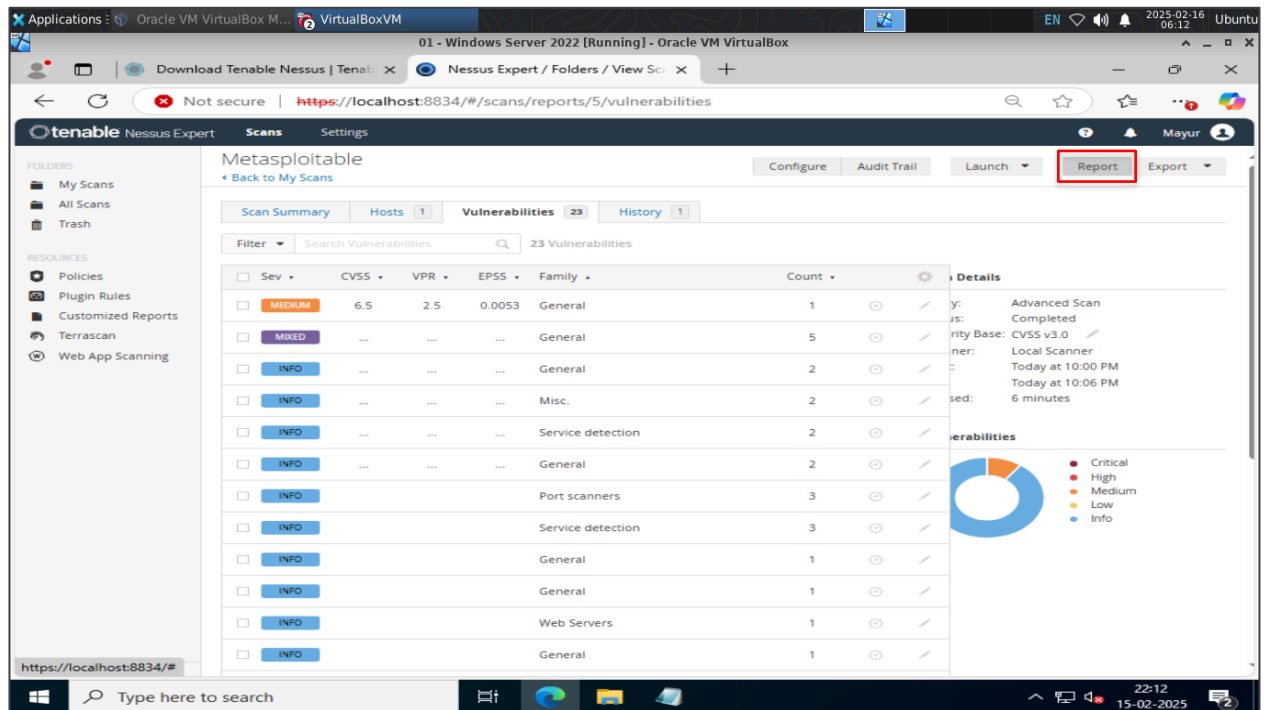
Scan Details

- Policy: Advanced Scan
- Status: Completed
- Severity Base: CVSS v3.0
- Scanner: Local Scanner
- Start: Today at 10:00 PM
- End: Today at 10:06 PM
- Elapsed: 6 minutes

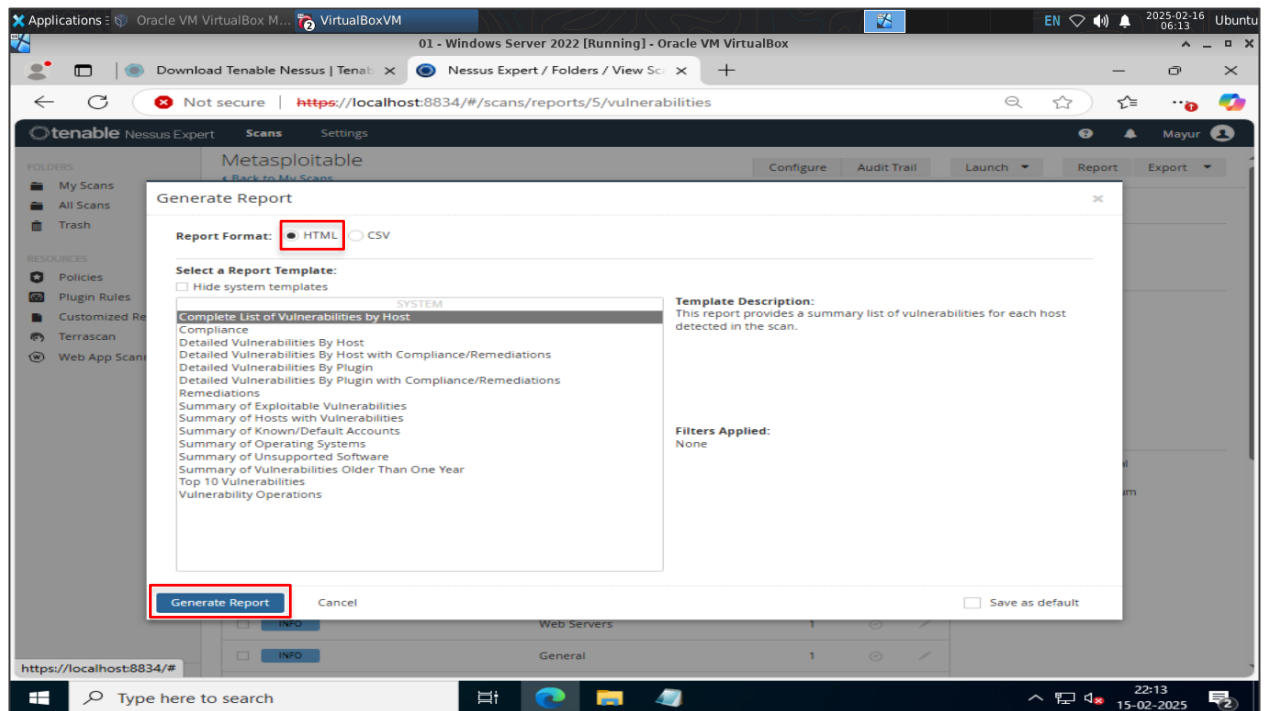
Vulnerabilities

- Critical
- High
- Medium
- Low
- Info

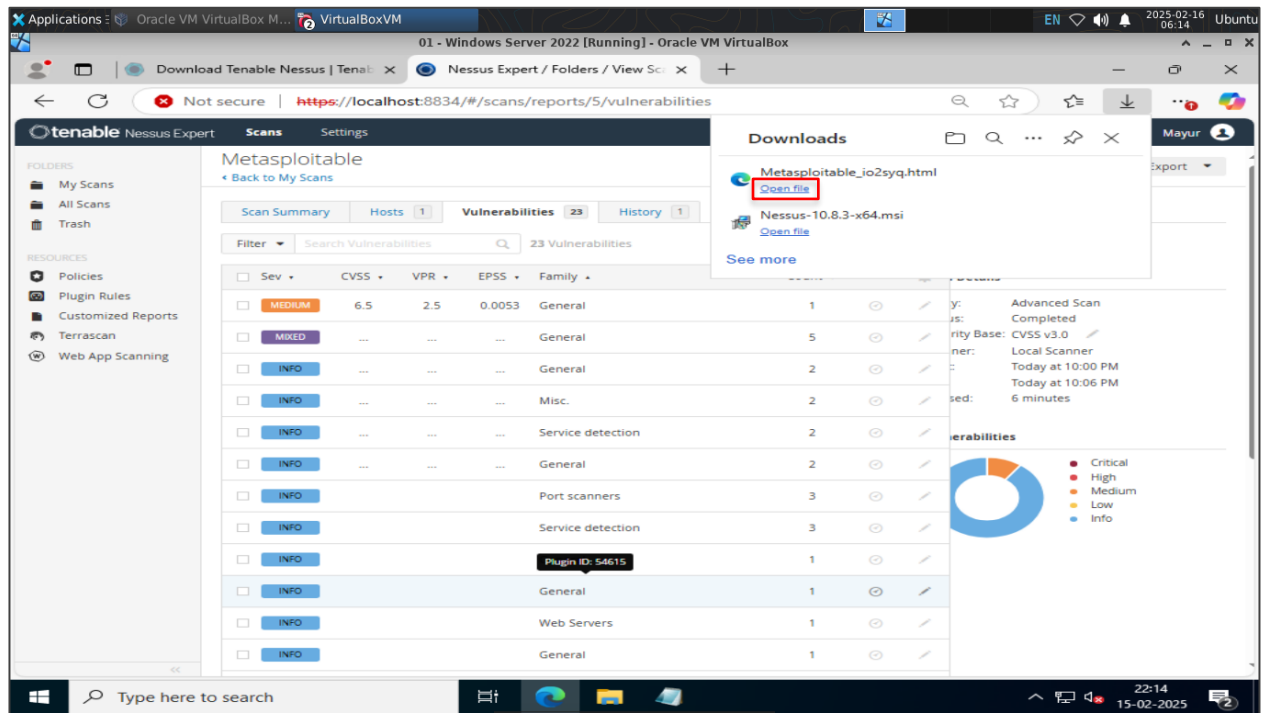
2.12 Click Report



2.13 Select HTML and click Generate Report



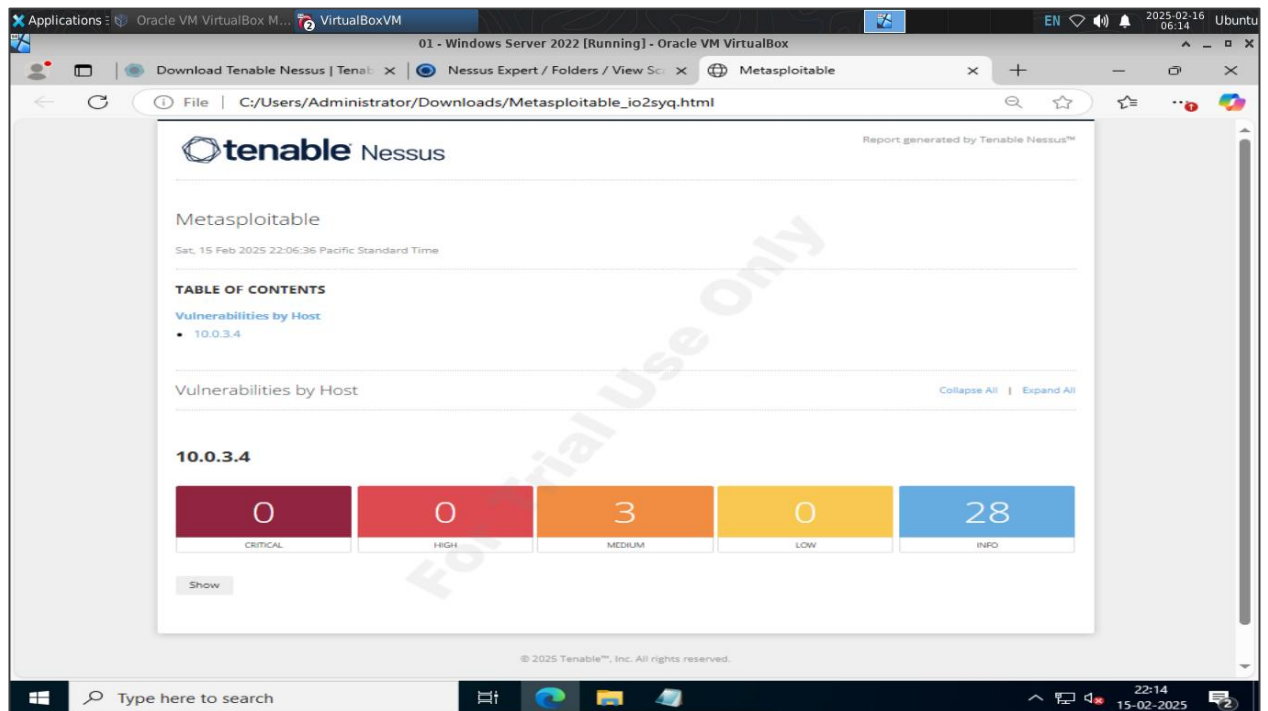
2.14 Click Open file



The screenshot shows the Nessus Expert web interface. The main panel displays a scan report for 'Metasploitable'. A download button is highlighted, and a download window is open showing the file 'Metasploitable_io2syq.html'. The download window also shows 'Nessus-10.8.3-x64.msi'.

Sev	CVSS	VPR	EPSS	Family	Count
MEDIUM	6.5	2.5	0.0053	General	1
MIXED	...	...	...	General	5
INFO	...	...	...	General	2
INFO	...	...	...	Misc.	2
INFO	...	...	...	Service detection	2
INFO	...	...	...	General	2
INFO	...	...	...	Port scanners	3
INFO	...	...	...	Service detection	3
INFO	...	...	...	Plugin ID: 54615	1
INFO	...	...	...	General	1
INFO	...	...	...	Web Servers	1
INFO	...	...	...	General	1

You can now view the downloaded scan report.



The screenshot shows the downloaded scan report file 'Metasploitable_io2syq.html'. The report is generated by Tenable Nessus. It includes a table of contents and a vulnerability summary for host 10.0.3.4.

TABLE OF CONTENTS

- Vulnerabilities by Host
 - 10.0.3.4

Vulnerabilities by Host

10.0.3.4

Critical	High	Medium	Low	Info
0	0	3	0	28

By following the above steps, you have successfully performed a non-credential scan using Nessus. This scan helps identify vulnerabilities on the target system without requiring login credentials. The scan results provide valuable insights into security weaknesses, categorized by severity, enabling you to assess risks and prioritize remediation efforts effectively.